University of Illinois at Chicago
Department of Chemical Engineering

CHE 381 Tuesday Sections EIB 128
9:00am-12:30pm or 1:00pm-4:30pm
Chemical Engineering Laboratory I Syllabus

Instructor:
Alan Zdunek
Office: 150 EIB
Email: zdunek@uic.edu
Phone: 312-996-4607

Required Course Text:
Unit Operations of Chemical Engineering
W.L. McCabe, J.C. Smith, P. Harriott
7th Edition, McGraw-Hill, 2005.

Office Hours:
The instructor and teaching assistant will be available during the lab period (9am-12:30pm and 1:00pm-4:30pm) to help with the laboratory experiments, analysis, and equipment maintenance. Instructor and teaching assistant will also be available to answer questions by email or in-person by appointment.

Introduction:
The laboratory component of an undergraduate education is critical to the professional success of scientists and engineers. CHE381 known as Unit Operations Laboratory I, provides students with a *hands-on opportunity to work with physical equipment in a laboratory environment*. Students, organized into small groups, will conduct experiments, and write a pre-lab plan and final lab report. Every two weeks, the lab teams will present the lab experiment to the next lab team. There will also be two safety assignments.

Lab Instruction
CHE 381 will be conducted **on-campus in EIB 128** following university guidelines for the safety and health of the students, instructor and TA's. There will be no remote access during the lab periods; all students must be present in-person. Lab teams in each section will work on different experiments during each lab experiment cycle. Each lab experiment cycle takes two weeks and will consist of a prelab period and a final lab period. Lab teams will perform 6 lab experiments by the end of the semester following this two-week lab cycle.

List of Experiments
Each team will perform the following six experiments in CHE381:

Humidification
Non-Newtonian Viscosity
Evaporation
Distillation
Membrane Separation
Fluid Flow in Pipe Systems

Lab Cycle

Two lab periods are allocated for each experiment as follows (See Figure 2 for Lab Semester Schedule)

Period 1 (Lab Prep): This lab period is devoted to becoming familiar with the equipment and preparing an experimental plan that will be carried out in Week 2. The team will review and understand the basic theory of the experiment, formulate a detailed step-by-step experimental procedure, and conduct a safety analysis on the equipment. A **Prelab Report** will be written by each team and is due the next day (end of day).

Period 2 (next week): This lab period is devoted to performing the experiment. The students assigned in Period 1 will come to the lab and perform the experiments and collect the data. After the experiments are done, the teams will analyze the results, formulate conclusions and recommendations, and investigate errors in measured variables. A **Final Lab Report** will be written by each team and is due the day before the next lab period (end of day).

Next Lab Period 1: Each lab team will be assigned a new lab experiment to perform. Lab teams will make an oral presentation to the next lab team doing the experiment.

Week 1:	Aug	26	Introductions; lab schedule	
Week 2:	Sept	2	**Safety Assignment I Due (BB Assignment)**	
Week 2		2	Pre-lab	**LAB 1**
Week 3		9	Laboratory	
Week 4:		16	Pre-lab	**LAB 2**
Week 5:		23	Laboratory	
Week 6:		30	Pre-lab	**LAB 3**
Week 7:	Oct	7	Laboratory	
Week 8:		14	Pre-lab	**LAB 4**
Week 9:		21	Laboratory	
Week 10:		28	Pre-lab	**LAB 5**
Week 11:	Nov	4	Laboratory	
Week 12:		11	Pre-lab	**LAB 6**
Week 13:		18	Laboratory	
Week 14:		26	Final Presentations ……EIB 224	
Week 15:	Dec	5	**Safety Assignment II Due No Lab Period**	
Week 16:	Dec	8-12	**FINAL EXAM WEEK** **No Lab Period**	

Figure 2. CHE 381 Laboratory Schedule for Tuesday AM or PM Section

Safety Assignments I and II: A safety assignment and quiz will be assigned through Blackboard assignment at the beginning of the semester to provide additional safety learning and comprehension. Students will also be required to complete a safety tutorial through the AIChE Academy Safety Training website as a Blackboard assignment.

The Blackboard website contains the most up-to-date manuals for carrying out the various unit operations experiments (pdf files). Useful information on each lab is provided in Blackboard to help aid in successfully preparing and carrying out the experiments, and to write the final report.

Chemical Safety in the Laboratory

First and foremost, safety is an integral component in all laboratory environments, and certainly of all experiments and industrial processes. The Chemical Engineering Laboratory Safety document is your guidance in this regard. You must a) read, sign and submit this document stating in effect that you have read, understood, and will comply with the regulations contained therein. The safety document must be completed before you begin any lab work in this course. It is a requirement that students in the lab take a proactive approach to safety and remind others of safety practices when necessary. The laboratory is an ideal place to develop safety awareness and skills; those who embrace the safety culture in the workplace will be highly regarded and rewarded professionally and financially in their future careers.

SAFETY GUIDELINES:
Leg Protection (No shorts) are required in laboratory.
Eating and drinking are not allowed in classrooms.
SAFETY GLASSES: Safety glasses must always be worn in the lab.
Compliance will be monitored by the instructor and TA during the lab period.
Non-compliance will be recorded.
Two (2) instances of non-compliance will result in a zero on the Individual Assessment grade.

Laboratory Access and Attendance

Student access to the Unit Operations Laboratory (128 EIB) will coincide with the scheduled class period; every Tuesday/Thursday from 9:00am to 12:30 pm (Morning sections) and 1:00pm – 4:30 pm (PM sections) throughout the semester. The nature of laboratory studies is such that they are time and activity intensive; ***please be on-time so that the laboratory can begin promptly*.** The experiments are designed to be performed in two class periods (pre-lab and lab). If additional time is needed, permission from the instructor is required to perform additional lab work.

Attendance Rules

Lab attendance will be recorded for each lab period and will be one of the criteria used in the individual assessment grade. Arriving 15 minutes or later will result in a late mark (L). Arriving after 50 minutes or later will result in an absent mark (A) for that lab period. Please notify the TA or instructor before class if you cannot make the lab period. One excused absence will be allowed and considered on an individual basis. **Two (2) or more unexcused absences or four (4) Late marks will result in a zero on the Individual Performance grade.**

Teamwork
Students will be organized into lab teams. Each team will conduct experiments, write pre-lab and final reports, and present their work bi-weekly to their colleagues and at the end of the semester. All team members are expected to participate jointly in planning and conducting the experiments, evaluating experimental data, analyzing results, writing reports and in the oral final presentation. The team should also decide on a team leader. Team member contributions will be tracked by each student filling out the Team Member Contribution form after each lab experiment (form is in Blackboard). The forms will be attached to the final lab report and will be used as part of the Individual Performance grade.

Academic Integrity -Student Community Standards
As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all your academic endeavors. When accusations of academic dishonesty occur, the Office of the Dean of Students investigates and adjudicates suspected violations of this student code. Unacceptable behavior includes cheating, unauthorized collaboration, fabrication or falsification, plagiarism, multiple submissions without instructor permission, using unauthorized study aids, coercion regarding grading or evaluation of coursework, and facilitating academic misconduct.

Submitting Assignments
By submitting your assignments for grading you acknowledge these terms, you declare that your work is solely your own or your lab team's work, and you promise that, unless authorized by the instructor or proctor, you have not communicated or collaborated with anyone in any way on your lab reports, presentations, or other lab assignments. Let's embrace what it means to be a UIC community member and together be committed to the values of integrity.

A Further Note on Plagiarism

Plagiarism is defined as "the unacknowledged inclusion of someone else's words, structure, ideas, or data." To avoid even the appearance of plagiarism, one must learn when and how to cite sources correctly. One helpful website is "Plagiarism.org". Plagiarism will not be tolerated and will result at a minimum in a zero for the report or presentation.